

Lab 1: Face Recognition and PCA Lab Report

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To implement this lab, I used Jupyter in collaboration with VSCode to make the code easy to read and fix. My code is split up into 9 distinct sections, all fulfilling at least one part of the lab instructions per section. I used Markdown and Python to code my project.

Results:

identification accuracy: 87.50%

Visualization Process:

average face



top eigenfaces

eigenface 1



eigenface 2



eigenface 3



eigenface 4



eigenface 5



eigenface 6



eigenface 7



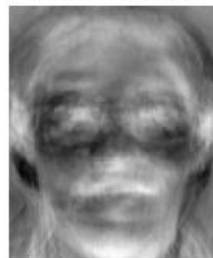
eigenface 8



eigenface 9



eigenface 10



bottom eigenfaces

eigenface 1



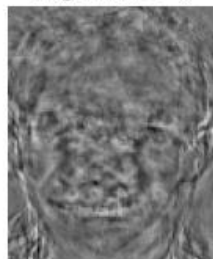
eigenface 2



eigenface 3



eigenface 4



eigenface 5



eigenface 6



eigenface 7



eigenface 8



eigenface 9



eigenface 10



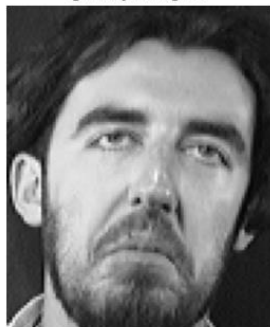
Random Images (varies each time code is ran):

subject 11 gallery images

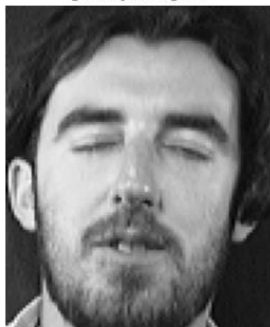
gallery image 7



gallery image 8



gallery image 9



gallery image 10



subject 3 test images

test image 1

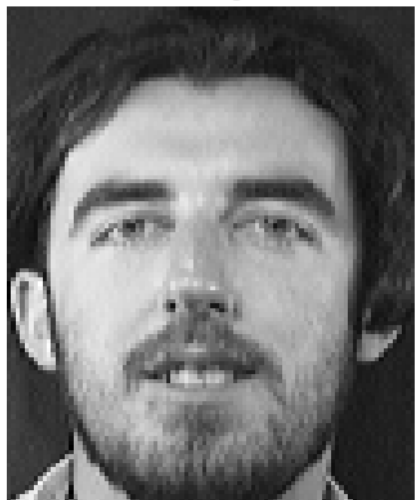


test image 2

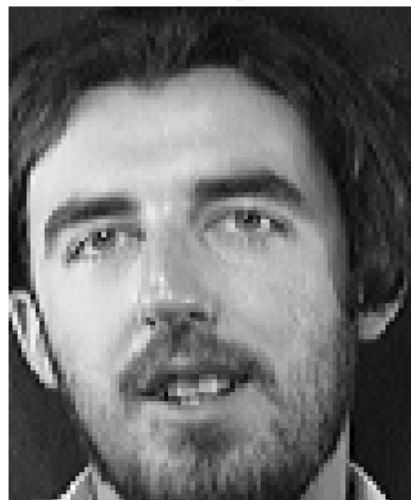


subject 11 test images

test image 1



test image 2



subject 3 gallery images

gallery image 7



gallery image 8



gallery image 9



gallery image 10

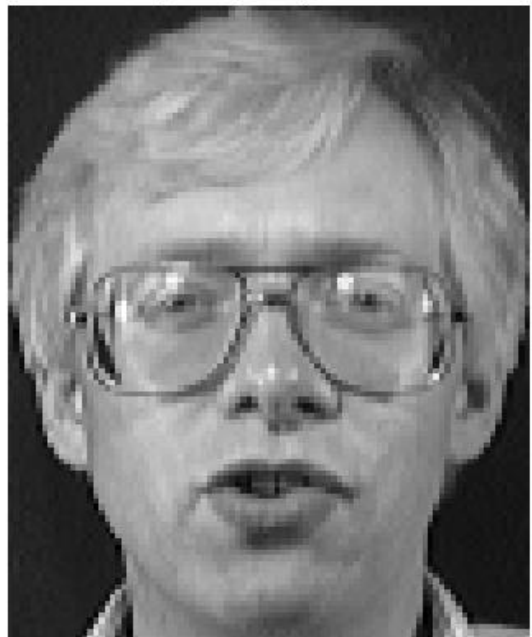


Matches:

test subject 2
(best match)



gallery subject 2



test subject 2
(best match)



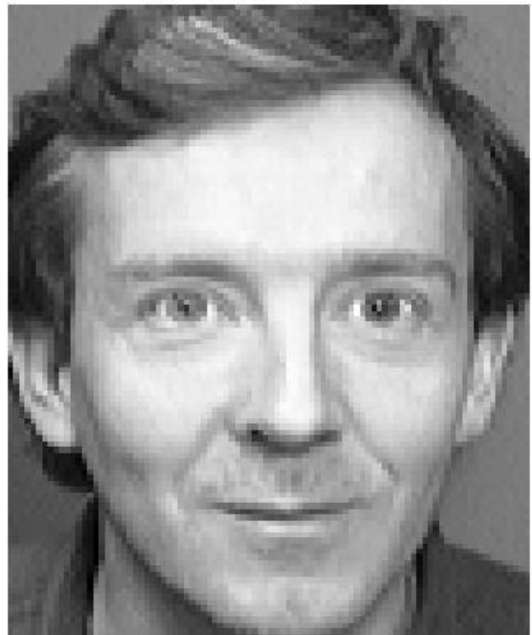
gallery subject 2



test subject 3
(best match)



gallery subject 3



test subject 3
(best match)



gallery subject 3



test subject 4
(best match)



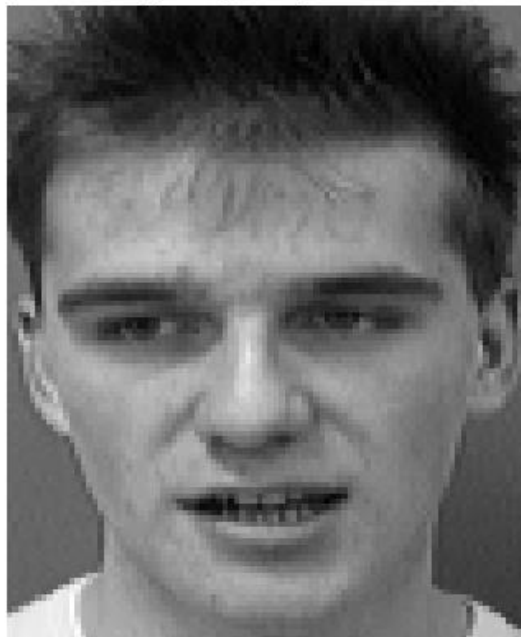
gallery subject 4



test subject 1
(worst match)



gallery subject 30



test subject 1
(worst match)



gallery subject 32



test subject 15
(worst match)



gallery subject 2



test subject 17
(worst match)



gallery subject 1



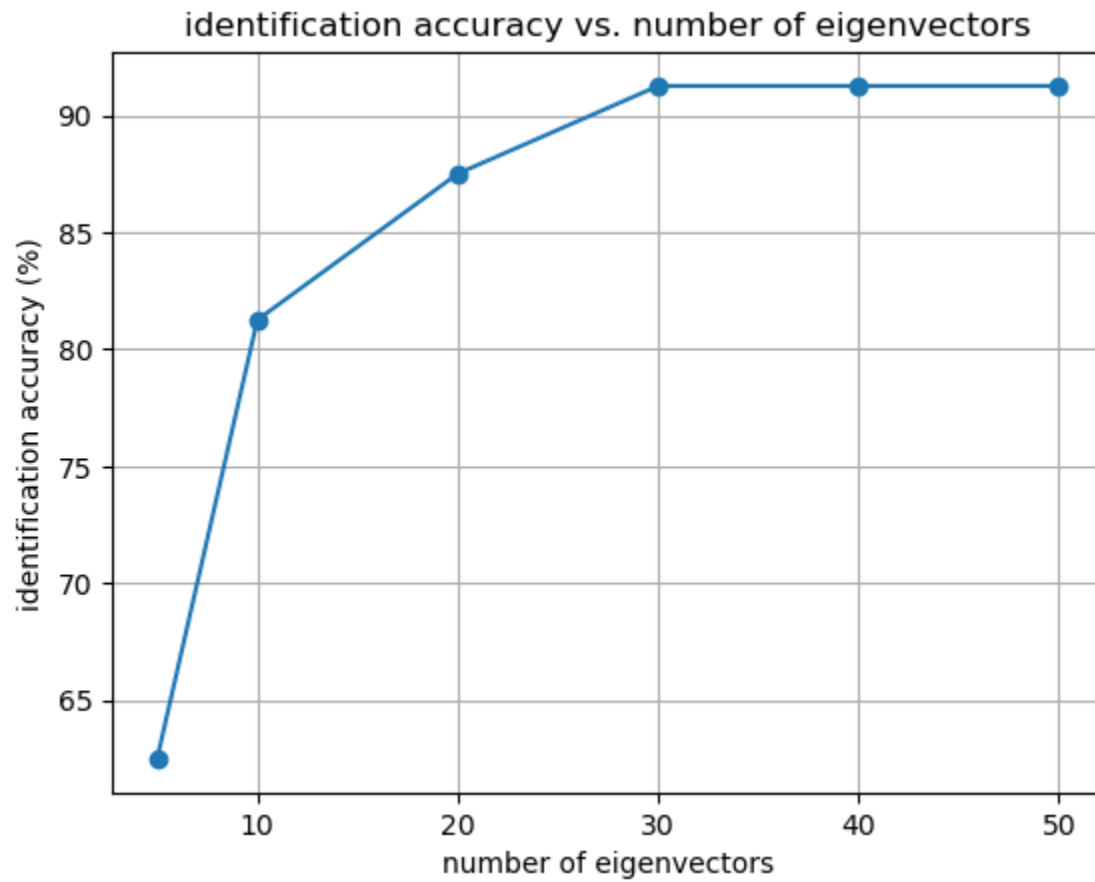
test subject 17
(worst match)



gallery subject 24



Eigenvalue Counts:



new split identification accuracy: 88.75%

There is no significant difference in the accuracy between C and A. This is possibly because the model is a robust model, and it is able to handle different configurations of images. The accuracy goes up slightly, meaning that the training of the model worked correctly.